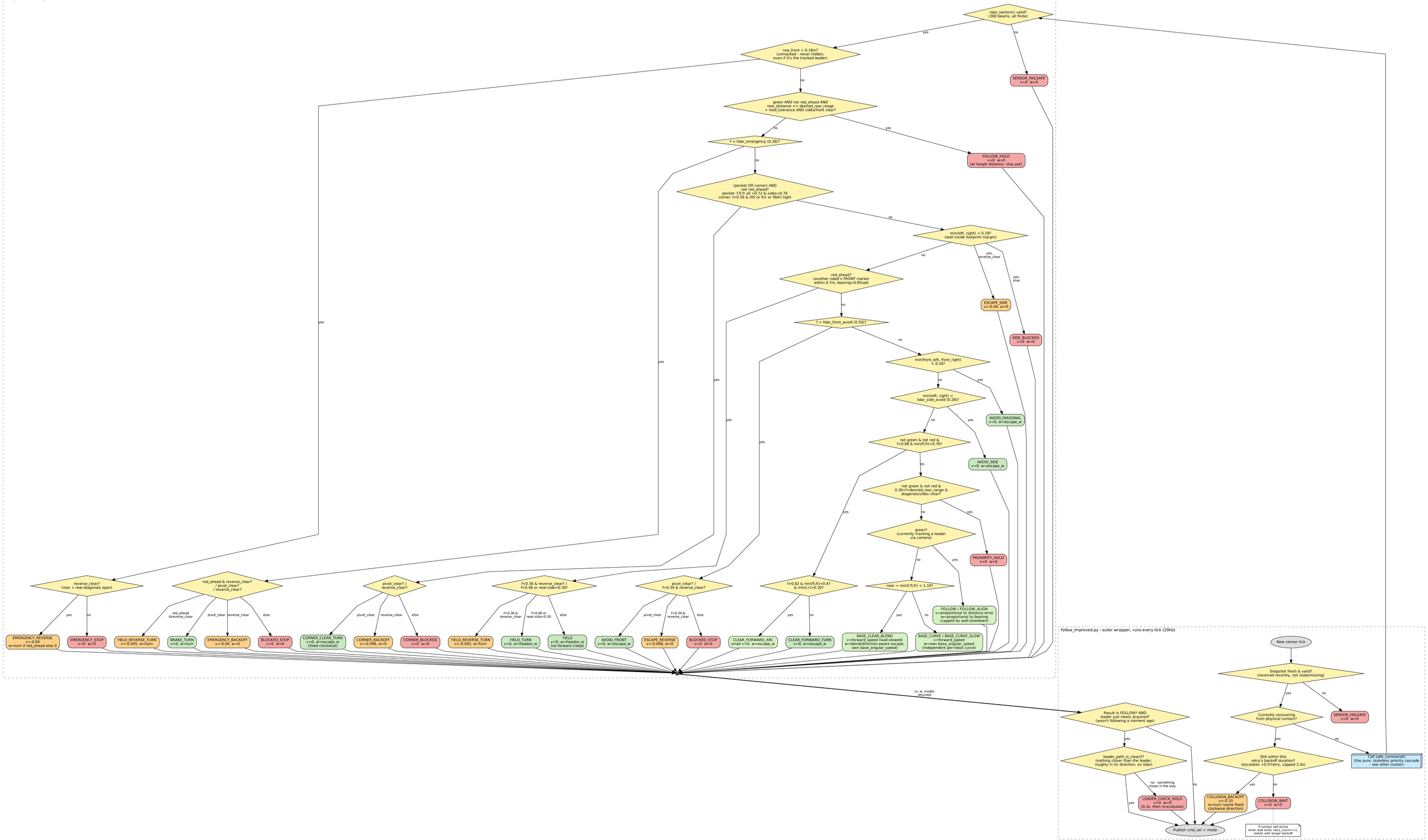
safe_logic.py - safe_command(): stateless, re-evaluated fresh every call, top-to-bottom priority
lidar_sectors() valid?
(360 beams, all finite)
yes
no
raw_front < 0.18m?
(unmasked - never hidden,
even if it's the tracked leader)
SENSOR_FAILSAFE
v=0 w=0
green AND not red_ahead AND
rear_distance <= desired_rear_range
+ hold_tolerance AND sides/front clear?
f < lidar_emergency (0.28)?
FOLLOW_HOLD
v=0 w=0
(at target distance, stay put)
(pocket OR corner) AND
not red_ahead?
pocket: f,fl,fr all <0.72 & sides<0.78
corner: f<0.56 & (fl1 or fr/r or fl&fr) tight
min(left, right) < 0.197
(wall inside footprint margin)
yes,
reverse_clear
yes,
else
red_ahead?
(another robot's FRONT marker
within 0.7m, bearing<0.85rad)
ESCAPE_SIDE
v=-0.04, w=0
f < lidar_front_avoid (0.50)?
SIDE_BLOCKED
v=0 w=0
min(front_left, front_right)
< 0.34?
min(left, right) <
lidar_side_avoid (0.28)?
AVOID_DIAGONAL
v=0, w=escape_w
not green & not red &
f<0.88 & min(fl,fr)<0.767
AVOID_SIDE
v=0, w=escape_w
not green & not red &
0.36<f<desired_rear_range &
diagonals/sides clear?
PROXIMITY_HOLD
v=0 w=0
green?
(currently tracking a leader
via camera)
near = min(f,fl,fr) < 1.107
FOLLOW / FOLLOW_ALIGN
v=proportional to distance error
w=proportional to bearing
(capped by wall-slowdown)
f>0.62 & min(f,fl,fr)>0.47
& min(l,r)>0.307
BASE_CURVE / BASE_CURVE_SLOW
v=forward_speed
w=own base_angular_speed
(independent per-robot curve)
BASE_CLEAR_BLEND
v=forward_speed (wall-slowed)
w=blend(direction-aware escape,
own base_angular_speed)
reverse_clear?
(rear + rear-diagonals open)
red_ahead & reverse_clear?
/ pivot_clear?
/ reverse_clear?
pivot_clear? /
reverse_clear?
f<0.36 & reverse_clear? /
f<0.48 or near-side<0.307
pivot_clear? /
f<0.34 & reverse_clear?
CLEAR_FORWARD_ARC
small v>0, w=escape_w
CLEAR_FORWARD_TURN
v=0, w=escape_w
EMERGENCY_REVERSE
v=-0.04
w=turn if red_ahead else 0
EMERGENCY_STOP
v=0 w=0
YIELD_REVERSE_TURN
v=-0.045, w=turn
BRAKE_TURN
v=0, w=turn
EMERGENCY_BACKOFF
v=-0.04, w=0
BLOCKED_STOP
v=0 w=0
CORNER_CLEAR_TURN
v=0, w=escape_w
(fixed clockwise)
CORNER_BACKOFF
v=-0.048, w=0
CORNER_BLOCKED
v=0 w=0
YIELD_REVERSE_TURN
v=-0.045, w=turn
YIELD_TURN
v=0, w=fheadon_w
YIELD
v=0, w=headon_w
(no forward creep)
AVOID_FRONT
v=0, w=escape_w
ESCAPE_REVERSE
v=-0.048, w=0
BLOCKED_STOP
v=0 w=0
(v, w, mode)
returned
follow_improved.py - outer wrapper, runs every tick (20Hz)
New sensor tick
Snapshot fresh & valid?
(received recently, not stale/missing)
SENSOR_FAILSAFE
v=0 w=0
Currently recovering
from physical contact?
Still within this
retry's backoff duration?
(escalates +0.5*retry, capped 2.0s)
COLLISION_BACKOFF
v=-0.10
w=turn (same fixed
clockwise direction)
COLLISION_WAIT
v=0 w=0
If contact still active
when wait ends: retry_count+=1,
restart with longer backoff
Call safe_command()
(the pure, stateless priority cascade
- see other cluster)
Result is FOLLOW* AND
leader just newly acquired?
(wasn't following a moment ago)
leader_path_is_clear()?
(nothing closer than the leader,
roughly in its direction, on lidar)
no - something
closer in the way
LEADER_CHECK_HOLD
v=0 w=0
(0.3s, then re-evaluate)
Publish cmd_vel + mode